

Example

$H_0: \mu = 10$

$H_1: \mu \neq 10$

```
mu0 <- 10 # null hypothesis
s <- 3 # sample stdev
n <- 20
xbar <- 8.5 # sample mean

stderr <- s / sqrt(n)
(t <- (xbar - mu0) / stderr)
```

```
## [1] -2.236068
```

```
(pvalue <- 2 * (1 - pt(abs(t), n - 1)))
```

```
## [1] 0.03754055
```

```
alpha <- 0.05
```

```
pvalue < alpha
```

```
## [1] TRUE
```

Reject H_0 at $\alpha = 0.05$

Sample

```
sample.mean <- replicate(n = 100, expr = {
  mean(rnorm(n=20, mean=10, sd=3))
})
summary(sample.mean)
```

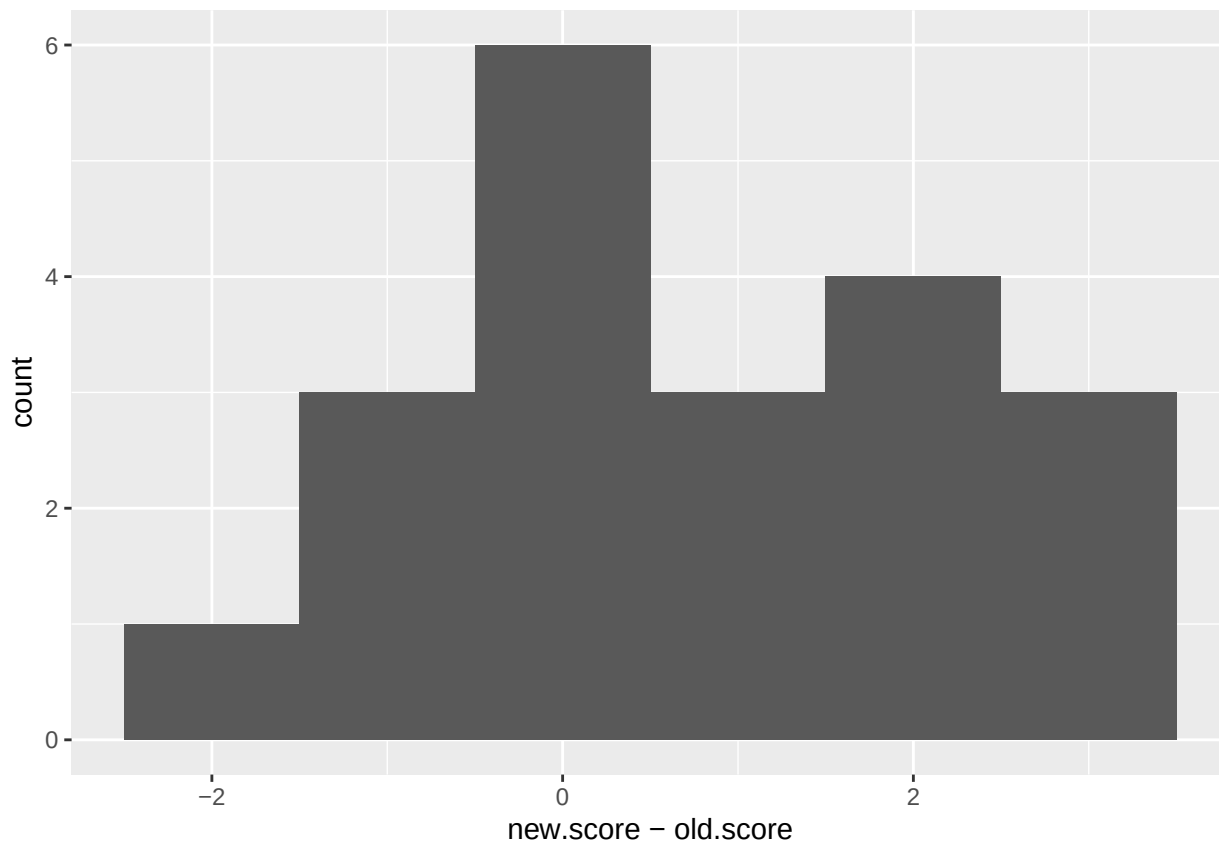
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   8.285   9.596   9.909   9.914  10.292  11.085
```

Scores Example

```
scores <- read.csv("https://raw.githubusercontent.com/njlytal/MATH314/main/scores.csv")
```

Plot Data

```
library(ggplot2)
ggplot(
  data=scores,
  aes(x = new.score - old.score)
) + geom_histogram(bins = 6)
```



Confidence Interval

```
xdiff <- mean(scores$new.score - scores$old.score)
n <- min(length(scores$new.score), length(scores$old))
sdiff <- sd(scores$new.score - scores$old.score)
alpha <- 0.05
```

Margin of Error

Assuming two-tailed test

```
tstar <- qt(1 - alpha / 2, n - 1)
stderr <- sdiff / sqrt(n)
margin_err <- tstar * stderr
(ci <- c(xdiff - margin_err, xdiff + margin_err))
```

```
## [1] 0.05623776 1.44376224
```

Hypothesis Test

$H_0: \mu_{\text{diff}} = 0$

$H_1: \mu_{\text{diff}} \neq 0$

```
mu0 <- 0 # null hypothesis value
(t <- (xdiff - mu0) / stderr)
```

```
## [1] 2.262689
```

```
(pvalue <- 2 * (1 - pt(abs(t), n - 1)))
```

```
## [1] 0.03556303
```

```
pvalue < alpha
```

```
## [1] TRUE
```